

Telling the Good from the Bad and the Ugly: How to Evaluate Backtested Investment Strategies

A growing share of the world's trading activity is generated by algorithmic investment strategies. Algorithms require development, backtesting, and investors that assume the initial performance risk. Evaluating the likelihood that backtested strategies will maintain their risk return profile in the future is an endeavor that requires experience and insight. In this paper, we describe a practical, non-technical approach to evaluating backtests that should help investors weed out those strategies the least likely to be profitable once launched in the marketplace. We also try to provide insight into why for most experienced investors the details of the statistical methods used to develop backtests are not the most critical immediate considerations when evaluating a potential investment.

BACKTESTS ARE EVERYWHERE

A growing fraction of all trades executed in listed securities around the world is driven by computer algorithms. As of 2009, high-frequency trading firms accounted for over 60% of all US equity trading volumes. Other markets such as fixed income, futures and currencies, also are seeing rising proportions of computer-generated trades.

Behind these algorithms lie both product sponsors who design and sell algorithmic investment products, and investors who buy these products. Regardless of whom the sponsors and investors are, the marketing pitch behind these products involves at the minimum a presentation of the hypothetical performance of the proposed strategy using historical price data, also known as a backtest. Our objective in this paper is to propose an approach to backtests that is borne from years of experience by your author in evaluating, managing and investing into algorithmic strategies. We will start illustrating our discussion with a simple backtest example. We then build on that example to demonstrate how investors should generally proceed to assess whether a backtested strategy is worth considering as an investment.

A BACKTEST EXAMPLE

There is an old saying amongst investors that “no one has ever seen a bad backtest”. Invariably, the teams that design investment products use their expertise to devise strategies with appealing performance. Poorly performing strategies are discarded or optimized to create the final product.

Let's look at a concrete backtest example. The date is 31 December 1999. Table 1 shows the past ten years of our example strategy's performance.

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Table 1: Strategy Backtest Example	
Length of backtest	10 calendar years
Annual returns	13.3%
Volatility	9.6%
Maximum drawdown	-9.9%
Sharpe ratio (@ 1% risk free rate)	1.3

Regardless of what the particular strategy is that produces these results, most investors will recognize that a ten year period is a short time in the markets. Much has been written recently about the lost decade of investment returns. So let's increase the backtest period to 20 and 30 years. The results are shown in Table 2.

Table 2: Strategy Backtest with Longer Time Frames				
Length of backtest	10 years	20 years	30 years	20 years – All out of sample
Annual returns	10.3%	15.0%	17.7%	19.9%
Volatility	9.6%	10.6%	10.1%	10.4%
Maximum drawdown	-9.9%	-10.6%	-10.6%	-10.6%
Sharpe ratio (@ 1% risk free rate)	1.3	1.3	1.7	1.8

Note that the strategy was designed using an initial 10 years of price data in Table 1, which is known as the “in sample” data. Table 2 incorporates the in-sample data as well as an additional ten and then twenty years of historical price data. The final column is based on twenty years of daily data entirely outside of the ten-year in-sample window, which is commonly referred to as “out of sample” testing. In this case our out-of-sample test ranged from 2 January 1970 through 29 December 1989. Note that the annual returns as well as the Sharpe ratios seem to improve with the length of the backtest time-frame, which is unusual although a generally positive sign.

Before reading further, think about what you would do if presented with the strategy above as an investment opportunity. What questions would you ask the sponsor?

Now that you've pondered these questions, let's first describe this strategy.

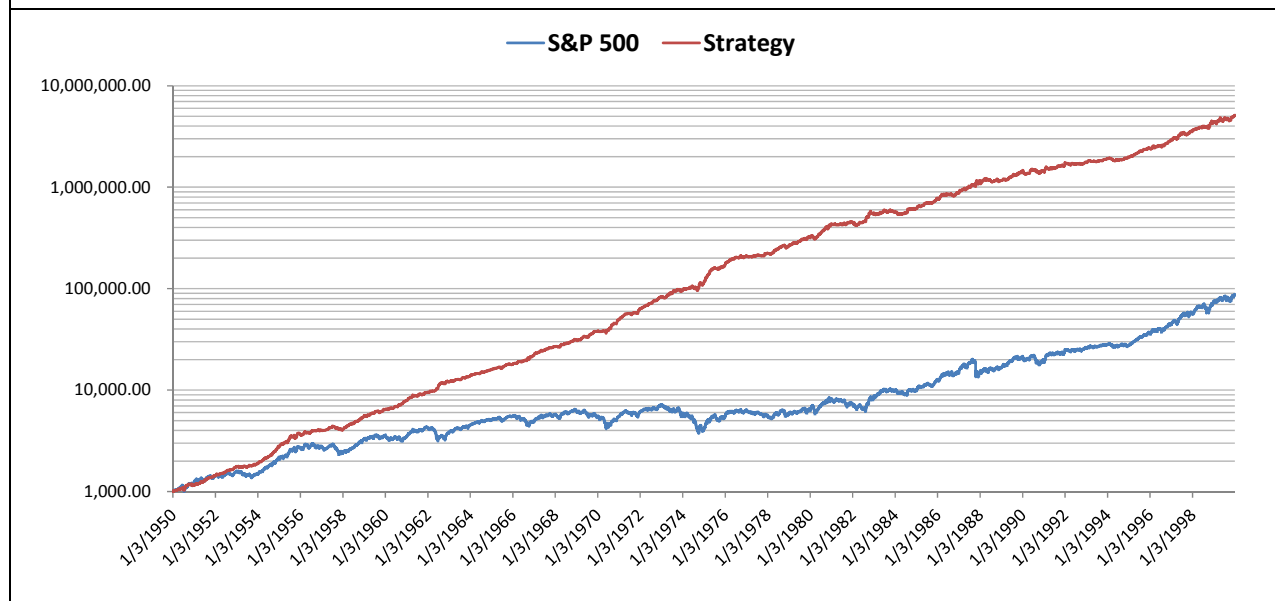
This is in fact a simple momentum strategy. The only security in the portfolio is the S&P 500 index (symbol ^GSPC). The strategy consists in buying the S&P 500 at market close if its performance for the day was positive, and selling the entire position otherwise. Trading costs were neglected.

Chart 1 displays the growth of the S&P 500 and the strategy between January 1950 and December 1999. In that fifty year period, the strategy returned 18.6% annually with a Sharpe ratio of 1.9 and a maximum drawdown of 10.6%. In the same period, the S&P 500 returned 9.4% with a Sharpe of 0.6 and a maximum drawdown of 48%.

What do these numbers tell us? And what are the key concerns an investor should raise?

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Chart 1: Growth of \$1000 Invested on 3 January 1950



In this case, we understand perfectly clearly what the strategy intends to do. It sells its S&P 500 position on down days, and buys it back at the end of up days. It is a short-term trend-following strategy. The main consideration for an investor therefore does not revolve around the length of the backtest, whether in-sample and out of sample data were used, or how many tests the sponsor performed before finalizing the investment approach. These questions have very little bearing on whether the strategy will make money going forward. The most relevant question is to understand why this strategy should make money at all going forward.

As a matter of opinion, your author would like to suggest that the backtest above is meaningless. It really does not matter whether it includes a long time period of fifty years, or whether proper statistical methods were. The real bottom line is that there is no particular economic phenomenon that is captured by a one-day momentum strategy applied to the S&P 500 index. Certainly chart 1 demonstrates that for 50 years starting in 1950, the numbers seemed to work. However were these numbers truly achievable during that time period? Until computers became prevalent and connected us all to the world's information, getting a stock quote from a broker required a phone call that may or may not get through immediately. It required the broker to check files, perhaps make calls and get back to the investor. Trading costs were usually charged as fixed commissions per share, and the S&P 500 index could not be bought and sold efficiently on market close. It is possible that had information been as easily accessible since 1950 as it is today, and trading as efficient, many market participants would have spotted this strategy. The opportunity to profit from the approach would have then rapidly been arbitrated away.

For these reasons, the strategy depicted in chart 1 from 1950 until at least the mid 1990's is simply a mirage in our opinion. The numbers are accurate, but what matters is the judgment call that the strategy would have been very difficult to execute operationally. Had it purported to reflect a particular investor behavior that could be counted on in the future, the strategy might have provided a reason to investigate how such behavior could be arbitrated today. Bereft as it is of any behavioral content, it lacks credibility although one cannot reach that conclusion purely from the numbers.

YOU CAN SEE IT BUT YOU CAN'T HAVE IT: THE MEANING OF OPERATIONAL EFFECTIVENESS

Our backtest example above touched upon the question of feasibility and operational capability. In 2013, an investor at home with a laptop and a high-speed connection can gain access to real-time price information, check the profit of the S&P 500 as of a few minutes or seconds before markets close, and execute the strategy proposed above before markets

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close. This was not the case ten years ago.

An investment proposal based on a backtest raises a host of operational questions. Is the product sponsor able to execute the investment with minimal difference between an investor's account and what the algorithm indicates? A strategy that trades once a week or once a month is much easier to implement than a day-trading strategy or even one that trades every day. The U.S. equity markets rise or fall over one percent a day on average for the past many years. Emerging market equities habitually fluctuate over two percent a day. A strategy that trades every day may in fact make most of its profits on a small number of days each month. If the product sponsor experiences significant challenges in its ability to trade the algorithm with consistently minimal tracking error, the long-term returns to investors are likely to suffer, even if the backtest was of exemplary quality and the ongoing algorithm shows some profitability.

Experienced investors will usually give thought to a product sponsor's operations before doing much review of any backtest. A perfect backtest that is unlikely to be executed without significant tracking error is a mirage. Investors should ask to see the sponsor's existing capital investment into the strategy and check that the trades and ongoing performance match the advertised algorithm over a reasonable time period. That check is likely to surface pertinent operational questions that may be hard to formulate from a review of the backtested numbers alone.

In other words, reviewing a backtest is not simply a question of gazing at pro-forma performance numbers and wondering about data samples. At least as important is developing confidence that the strategy can be executed effectively by the sponsor.

The questions in that regard are commonsensical, and investors should not back away from asking them. Who will trade the strategy? Is that one person or a team? What happens if the principal trader goes on holiday, or gets sick? Where is the trading desk? What markets does the strategy trade and what are the implications – for instance some markets trade continuously while others open and close at various times in different countries. What other strategies is the trading team working on besides that presented in the backtest? What procedures are in place to deal with natural disasters or emergencies? The list goes on...

An example of being able to see an opportunity that cannot be captured is the spread between futures and cash markets. Because equity futures markets are open when the U.S. equity markets are closed, the price of corresponding securities in the futures and cash market will often seem to offer an arbitrage opportunity when equity markets open at 9:30 in the morning New York time. A backtest of such a strategy could look appealing. However, those types of arbitrages in practice are really only available to well established institutions with the technology and reach to capture price differentials within seconds of market opening. The average market participant has no real opportunity to capture these arbitrages on a consistent basis.

A minimal understanding of how markets work and of the operational capabilities of a product provider can suffice to avoid even considering a backtest.

PSYCHOLOGY

A backtest typically will include an analysis of the strategy's historical drawdowns, which are the peak-to-trough losses that mark the evolution of an investment strategy. It is important to realize that all investment strategies with a reliable mark-to-market are in a drawdown most of the time. It is only occasionally that the strategy reaches a new all-time high.

The question of how deep drawdowns are and how long they last is important for two reasons. First, it indicates to investors what losses have happened in the past and will happen again under similar market conditions. Second, it is a cue for investors to assess the psychological ability of the money manager to act under the pressure that comes with losing money.

A strategy with a ten percent average return and Sharpe ratio of one has a volatility of about ten percent – neglecting for now the risk free rate. At some point, that strategy is likely to experience a loss between ten and fifteen percent

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corresponding to a two standard deviation event.

There is a very significant difference between looking at a backtest on paper and trading a strategy where losses can accumulate for days or weeks, and where recovering from drawdowns can take months. In the face of adverse markets and nervous clients, a money manager needs to remain emotionally stable. The pressure from clients to “fix it” or “do something about it”, the relentless siren song with the refrain “this time it’s different” that sows seeds of reasonable doubt in the strategy’s soundness going forward, are very powerful forces. Under these circumstances, a manager needs the maturity and confidence to either hold the course or make measured adjustments to the strategy not overly influenced by recent performance. Above all, the management team needs to continue executing the strategy on a daily basis. If the strategy offers investors the ability to redeem frequently, investors may leave the strategy and move on. The money manager does not have that option. Emotional maturity and stability are key requisites for successfully getting through an even moderate drawdown.

Although a backtest can appear mostly as a list of numbers, the team that is executing, monitoring and doing research and development work on the strategy is human. If a key member of the management team does not seem to the investor like the type of person who would do well under high pressure, or has never actually lived through a significant drawdown, an investor would do well to ponder whether further review of the backtest is an exercise in wishful thinking.

UNDERSTANDING THE STRENGTHS AND WEAKNESS OF THE STRATEGY’S INVESTMENT STYLE

If we can get past the issues of operational effectiveness, maturity and psychological stability, the next step in reviewing a backtested strategy is to get a clear understanding from the product sponsor of what the strategy is designed to do independently from the numbers or the tactics used. That conversation usually feels like a tug-of-war. The investor’s objective is to understand the economic rationale behind the strategy, under what market conditions should it do well and why, and in what markets it will lose money and why, without reference to algorithms or numbers. The product sponsor will naturally want to refer to charts, graphs and algorithmic techniques, all of which are a means to an end but not the insight being sought.

A sponsor who is unable to articulate simply and clearly what the economic rationale of the strategy is, probably does not have a viable long-term strategy. A money manager may try to explain *what* the strategy does, for instance stating ‘we arbitrage short term deviations in value between the Swiss franc and the U.S. dollar, then apply the same technique to other currency pairs’ – this is a recent conversation between your author and a money manager. The real question is why the manager believes such arbitrage opportunities exist in the first place. Without a convincing explanation, a backtest may simply be a collection of lucky numbers.

As a case in point, let’s revisit our initial trading example in chart 1. We saw that between January 1950 and December 1999, a fifty year period, the strategy returned 18.6% annually with a Sharpe ratio of 1.9 and a maximum drawdown of 6.8%.

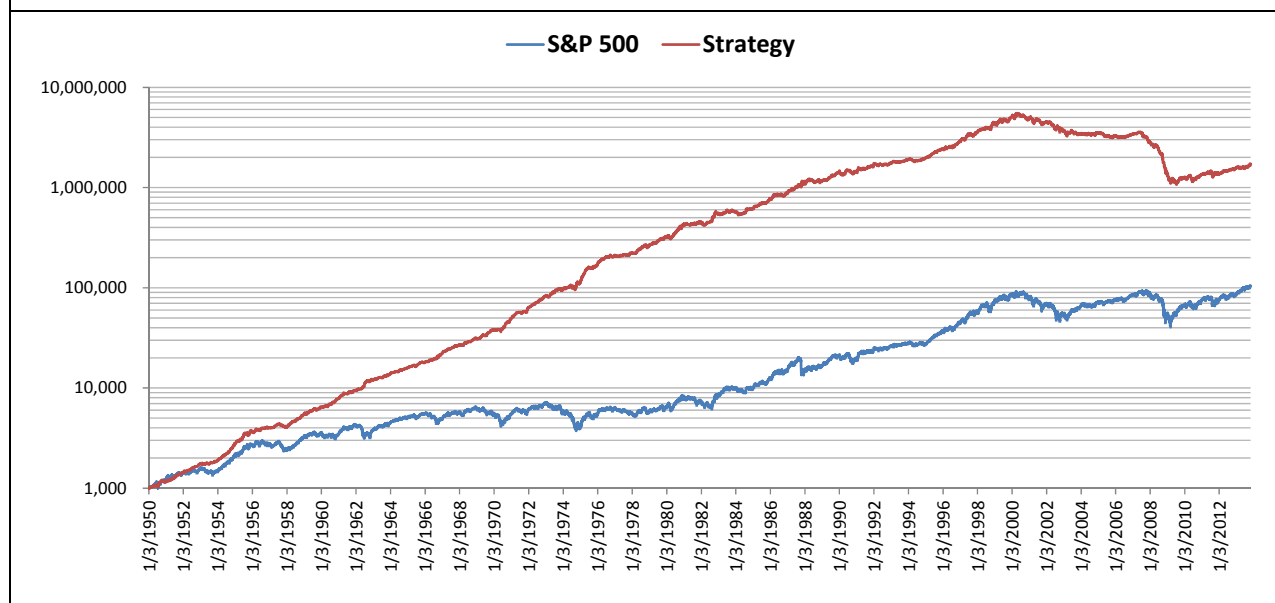
For the following ten year period starting in January 2000, this exact same strategy stopped working altogether. As shown in chart 2, it lost 13% annually for the ten years between January 2000 and December 2009. So what happened? Perhaps the wider availability of price data, the advent of computer trading, the rise of tracking ETFs that made trading the 500 stocks of the S&P 500 index more efficient all played a role. From our point of view, this strategy had good numerical results through 1999 but no clear reason of being. Thoughtful investors would have stayed away in 1999 as they should today, despite the 8.8% annual return from March 2009 through October 2013.

Understanding best and worst environments for a strategy is easier if it is designed as one of a number of well-established investment styles. Long-only strategies tend to buy and hold securities. Other styles include relative value, event-driven, equity hedge and macro investing, a brief definition for each of which is provided at the end of this paper. While each investment style can be implemented using quantitative algorithms, the most common algorithmic strategies employ a combination of arbitrage, trend-following or momentum, and global macro approaches.

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Each of these styles has inherent strengths and weaknesses which are important guideposts in a thoughtful review. An arbitrage opportunity rarely lasts for long periods. It may disappear as more market participants catch on and execute a similar trade, or simply as a result of regulatory or other economic forces. Understanding why and how one would expect to recognize that the trade opportunity has disappeared from the markets is of paramount importance.

Chart 2: Growth of \$1000 Invested
3 January 1950 to 23 October 2013



Strategies based on mean-reversion, namely the idea that a temporary divergence between a particular metric of two securities – usually their price or volatility – will eventually disappear as the relationship between these securities reverts to its usual state, may only be able to generate infrequent trading opportunities. Past results in such a strategy may be highly reliant on a few concentrated trades having properly worked out, one of many potential risks hidden behind the numbers.

Whether the economic environment going forward is likely to be as favorable as in the past for the specific strategy is a key question to come to grips with, before even considering any specific algorithm. As of this writing in late 2013, the most important systemic change facing investors is the direction of long term interest rates. There are many other important changes facing us all, such as the transformation of emerging economies into developed markets, the effort by various countries to diminish their dependence on the U.S. dollar, the upcoming end of the dependence of western nations on middle eastern energy suppliers to name but a few. All these changes are likely to transform investment markets over the next few years.

In summary, understanding what investor behavior or economic circumstance a strategy is intended to benefit from, what market regimes will be favorable and unfavorable, as well as developing a sense that markets going forward are not particularly adverse to the investment style, are both possible and necessary. Developing that appreciation should be done without any reference to the numbers or algorithm being presented as a backtest.

NO THANK YOU: OVERLY PARAMETRIZED ALGORITHMS NOT ACCEPTED HERE

Let's take stock of where we stand. Presented with an investment strategy backtest, we checked that the strategy sponsor has the operational capability to execute the strategy in the markets that will be traded. Those checks included a review of the team's own trading record for this strategy over a reasonable time interval. We have come to believe

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that the investment team is mature, experienced, and likely to perform under pressure when markets temporarily turn against the strategy, as they eventually will. Finally, we have developed an appreciation of why the overall approach can make money, as well as what market conditions are most and least favorable given the investment style.

That commonsensical process did not involve actually analyzing any backtest numbers or methodology. Neither did it require deep expert knowledge of the financial markets. It required being thoughtful and controlling the greed factor that naturally surfaces at the sight of a good-looking backtest. It has eliminated from consideration most wannabe quantitative money managers, as well as those strategies for which no convincing explanation of 'why it should work' is forthcoming.

We are now ready to examine the methodology of the backtest itself.

An initial concern with any backtest is whether the data used is sound. Financial data usually needs to be "scrubbed" to represent a reliable and usable data package. The corporate actions of companies are one of many reasons why this is necessary. For instance the payment of dividends creates an adjustment in share price that needs to be account for. Mergers, acquisitions, divestitures similarly impact stock price data. We will assume that the product sponsor uses verifiably reliable data since this is not the central issue of this discussion.

The central concern in evaluating backtested performance is what is known as overfitting the data. Nowadays computer power enables the development of complex quantitative strategies that can be optimized to deliver the best performance statistics. The single best question an investor can ask to assess the likelihood that a backtest overfits the data is the following: How many parameters drive this strategy?

This is the key question because it goes to the heart of what algorithmic strategies are good at. If market conditions repeat themselves exactly, an algorithmic strategy will faithfully deliver the performance that it did in the past under the same conditions.

The problem is that markets conditions never seem to be exactly replicated over time. Although equities experienced bull markets between 1996 and March 2000, and again starting in March 2009 through the present, markets in the late 1990's are quite different than today. Commodities are in a significant bear market today; interest rates are low and rising; a number of emerging markets of yesteryear have actually emerged; market participants commonly use computers to trade; credit markets in the U.S. have significantly changed since 2008, and so on. These differences are sufficient to change the behavior of investors and therefore of that of asset prices from one equity bull market to the next.

Consequently, an algorithm that is highly dependent on markets precisely repeating themselves is highly risky. The more numerous the parameters used to define a strategy's behavior, the higher the chance that performance is driven by specific interactions in the marketplace that may not repeat exactly. This is true whether the algorithm is tested on long or short time-frames as well as on so-called 'in-sample' or 'out-of-sample' data, a question we will address shortly.

Algorithmic strategies employ four categories of parameters which may overlap depending on the nature of the strategy.

1. Those that define the universe of securities in the strategy.

Some strategies constrain their security universe a priori. For instance there are legions of futures trading strategies focused solely on the S&P 500 E-Mini contract. Other strategies may include all listed large-capitalization equities in the United States, and then create a set of rules to determine which of these qualify for consideration on any given day. Momentum-based strategies are well suited to this type of approach.

2. Asset allocation rules that determine how capital is allocated across securities.

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Statistical arbitrage strategies track a statistically meaningful event, for instance a significant price spread between two securities relative to historical norms. When that spread is reached, the allocation rules would specify to short the high priced security and buy the low priced one. How much capital is allocated to each security also needs to be ruled on. Other related parameters include the length of time used to compute price volatility, momentum or other financial metrics.

3. Trading rules that trigger portfolio rebalancing or define trading events.

A balanced investment strategy will seek to rebalance its portfolio on a regular basis. The frequency of rebalancing is a key parameter that in this case triggers an assessment of the asset allocation rules. For a statistical arbitrage strategy, spread reaching specified widening and narrowing targets would trigger a trade. Trading rules also include assumptions on trading costs, slippage, rounding of shares, minimum portfolio size and various other practical issues that need to be decided upfront.

4. Risk control rules.

Risk control rules may be interwoven into asset allocation or trading rules, or may be independent from them. For instance stop losses can be applied at a security or portfolio level, regardless of the particular asset allocation at the time they are triggered. Leverage is sometimes managed on the basis of perceived macroeconomic opportunities. The severity of a strategy's drawdown is also often used to reduce leverage. Asset allocation rules then handle how the available capital gets allocated across individual securities.

Note that some strategies do not use certain parameters. An experienced investor will realize that the absence of a parameter can in itself be a parameter. For instance a strategy may not use leverage or individual security stop-losses. But these levers are available. There might be excellent reasons not to use them, such as tax or regulatory constraints. However it may also be that the strategy does not work as well when these are incorporated. A strategy may also use double smoothing techniques while another will not. A momentum strategy may include a minimum profitability threshold for a security to qualify for allocation, while another may not. Recognizing these implicit choices requires experience. One often finds that implicit choices can be as numerous as the explicit decisions represented by driving parameters.

As a rule of thumb, simpler strategies seem to perform better than complex ones. Simpler strategies by definition have fewer controlling parameters than complex strategies. Empirical evidence gathered over the course of many years suggests to your author that backtests with more than a dozen key parameters should be handled with circumspection. While a dozen seems like a large number, most strategies will get there fairly quickly across the four categories mentioned above. It becomes difficult to understand the impact of optimization across too many variables. It is then a challenge to understand why a trade is made, what market environments will be favorable or adverse to the strategy, and what really drives performance over time.

It is interesting to ponder the devil's advocate position on this complexity point. Consider discretionary strategies, i.e. strategies in which a portfolio management team, and not a computer program, makes allocation and trading decisions. How many parameters does a portfolio manager, let alone a team of professionals, use to make just one decision? The cumulative impact of years of experience, impressions about current market conditions, recent client input, corporate imperatives, team dynamics, personal biases, the state of one's health, all probably add up to hundreds if not thousands of individual inputs. No one can claim that a discretionary investment process is repeatable. A common saying amongst algorithmic money managers is that there is no bigger black box than the brain of a discretionary trader. A discretionary manager can put on a trade and justify it with a perfectly good explanation based on a number of considerations. An algorithmic strategy is constrained to repeating its same algorithm, and we have suggested that to be credible its governing parameters need to be few. Today, we will we trust a discretionary manager influenced by a multitude of sensory and cognitive inputs, but few experienced investors will trust an algorithm with a more than a few pre-defined parameters. It is conceivable that in the distant future artificial intelligence will be so advanced that algorithms will be easily designed that handle a great number of degrees of freedom with something that feels like investment savvy. As of today however, money Heaven is oozing with investor capital lost in investment strategies

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that tried to mimic neural networks, wave propagation theories and other natural systems. The time for these complex strategies has probably not yet come.

For these reasons, an ideal algorithmic strategy today involves only a handful of controlling parameters, supplemented perhaps by a few more variables that are necessary to define the trading approach but with respect to which the strategy is not materially sensitive.

For instance a diversified multi-asset class strategy is unlikely to be very sensitive to the addition of one or more securities to its universe. That sensitivity has ebbed away asymptotically under the impact of sufficient diversification. However, despite being diversified, such a strategy is likely to remain sensitive to the specific asset allocation process that defines how capital is allocated to each security in the universe. The parameters that define asset allocation are thus controlling parameters.

Let's revisit our original strategy example in chart 1. One parameter defined the universe of securities, in this case the S&P 500 index. The asset allocation rules, namely to buy when the security is profitable and to sell when it is unprofitable, create two more controlling parameters. Finally, the time window over which profitability is evaluated, one day in this case, is also a key ingredient. The strategy has four controlling parameters. The risk control rule is part of the asset allocation approach, namely to sell upon a loss. As a result, it is very easy to understand what this strategy will do. This does not mean that it is simple to understand why it should make money, as it remains a soulless set of rules.

Our initial example brings out a rather pernicious psychological effect of algorithmic strategies. It is easy for investors to be lulled into a sense of comfort once they understand how an algorithm works. If in December 1999 an investor were presented with the strategy in chart 1, the combination of feeling that the strategy is easy to grasp, that the track record is difficult to argue with, mixed perhaps with a touch of satisfaction at one's well-deserved good fortune in finding a promising money manager, might be enough to separate the investor from his or her money. While all these emotions are natural, the key question left unanswered was why the strategy should continue to make money. In fact we saw that it did not after December 1999.

We have now weeded out strategies that seemed credible but involved too many driving parameters. Note that we have done so without spending much time or effort in actually analyzing the backtest. We also have shown that having few parameters does not guarantee that the strategy will perform, regardless of what the backtest indicates.

BUILDING COMFORT WITH THE BACKTEST'S DRIVING PARAMETERS

Once a strategy passes the complexity screen, the easiest path to further qualify a backtest is to check the nature and level of the four parameter categories listed in the previous section. Regardless of the numbers in the backtest, the investor should ensure that these driving parameters are sensible.

A futures trading strategy will normally use a target risk level that defines how much leverage is employed. Larger risk limits enable higher leverage, hence more volatility and risk. A ten percent risk limit is fairly commonly used, although some product sponsors will offer twice that limit or more. Other strategies such as long/short equity, fixed income arbitrage and market neutral approaches also normally involve the use of leverage.

For these types of strategies it pays to keep in mind the old adage: *Bad trades hurt you, leverage will kill you.* Verifying that leverage limits are sensible is a priority. Equity market neutral strategies that employ one to three times leverage are reasonably common. Some have limits of five times leverage or more. An investor not comfortable with advertised leverage limits need look no further at the backtest results.

A dead giveaway that the backtest of a leveraged strategy is guilty of overfitting data can be found by checking its history of drawdowns. In general leverage amplifies returns as well as the risk of being caught levered in a suddenly adverse market. That situation will result in deep and sudden losses in comparison with the strategy's own usual drawdown pattern. In our experience, the absence of these sudden and large drawdowns in a leveraged backtest

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significantly increases the likelihood that the results are not repeatable.

The questions that investors should ask themselves when reviewing backtest parameters are legions. In large part they depend on the investment style of the strategy. Is the strategy concentrated or diversified? Do the rules that include or exclude securities make common sense? Are the asset allocation rules based on well-established financial metrics? Do risk-control rules make sense or do they feel so abrupt that it is difficult to develop an intuition in how they would work when market regimes shift?

These questions are not particularly different than those one would customarily ask of a discretionary manager. When investors quiz a discretionary manager on what will happen under various market scenarios, both parties know full well that future decisions by a human team will be subject to many influences at that time. The manager can give a general sense of what process the team is likely to follow, but the precise outcome cannot be certain. In computer-driven trading, what you see is what you get today and in the future. It is important for an investor to understand what the driving parameters are, what they mean, and why they are set at a particular level.

This approach will weed out strategies that while potentially profitable, are more aggressive than the investor is likely to be comfortable with. It does not necessarily require deep expertise in either finance or algorithm development. It does require commonsense and an investor that cares.

REVIEWING ECONOMICALLY-ROOTED STRATEGIES

We are now ready to analyze the methodology of the backtest that has survived our elimination process. Let's consider a new backtest example that will illustrate our discussion. We will use a simplistic trend-following strategy applied to the S&P 500 index is as follows:

1. **Securities universe:** S&P 500 index only
2. **Asset allocation rules:** 100 % allocation to cash or to the S&P 500 index
3. **Trading rules:**
 - a. The portfolio is rebalanced on the first business day of each month.
 - b. Allocate to the S&P 500 if its momentum is positive.
 - c. Momentum is assessed over a period of 11 months starting 12 months ago. Note that the most recent month is not included in the calculation.
4. **Risk control rules:** If momentum is negative when rebalancing, go to cash.

This investment style is easy for all to understand. It purports to take advantage of the momentum effect which is well documented in financial markets.

Now ask yourself what is the point of a backtest for a strategy rooted in an economic phenomenon that is documented across asset classes and decades of market data^{1 2}?

We understand that if the S&P 500 falls over many months, the strategy will initially take a loss and eventually move to cash. It will come back into the markets after it sees a rise in the S&P 500 over an eleven-month period, which implies that it will miss a potentially substantial part of an initial rebound, although it will also miss a potentially long period of continued losses when it moves to cash. On the other hand during a sustained equity bull market, we could get returns comparable to those of the S&P 500.

¹ M. Faber, April 2010. *Relative Strength Strategies for Investing*. Social Science Research Network.

² G. Antonacci, January 2013. *Risk Premia Harvesting Through Dual Momentum*. Social Science Research Network

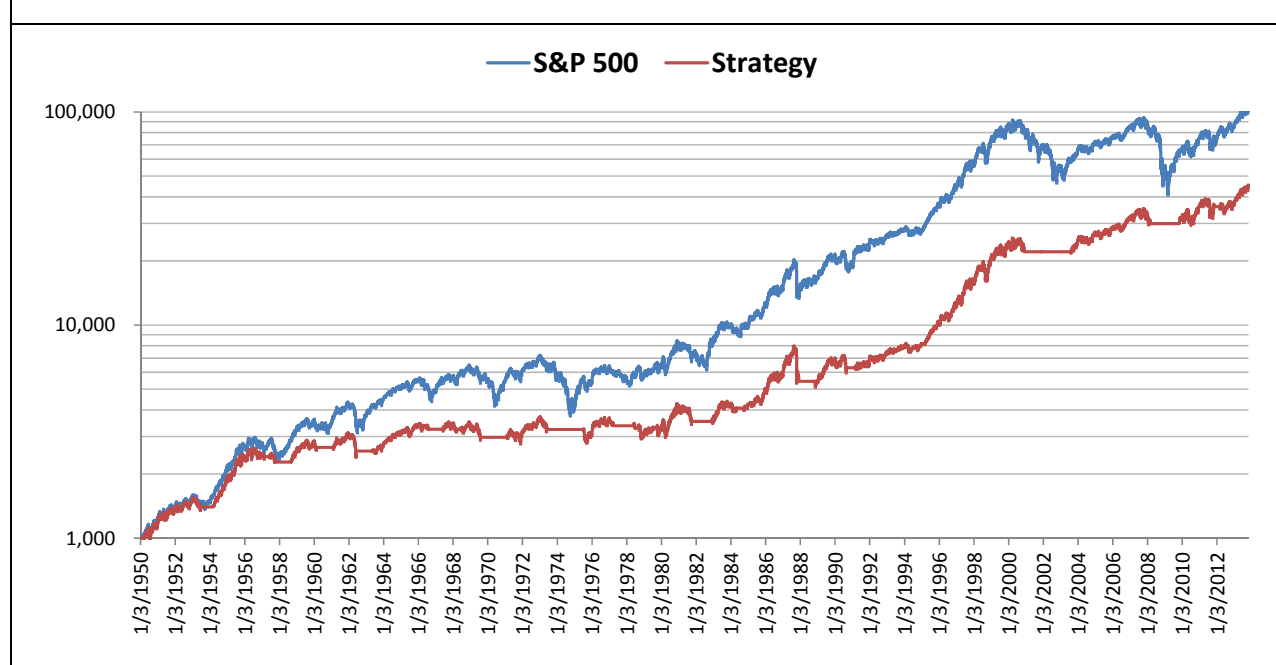
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What we need to know as investors is what returns characteristics we should expect through the various types of market cycles that equities do go through. The best backtest analysis is one that reaches far back into the past to capture the characteristics of the maximum drawdown this strategy would have experienced – length, depth, time to recovery from trough are typical metrics. It is also useful to isolate particular market regimes to understand the pattern of returns and losses during these periods.

The performance of the strategy is shown in Table 3. Its equity growth curve is displayed in chart 3.

Table 3: Strategy Backtest				
Length of backtest	Past 10 years	Past 20 years	Past 64 years	4 year bull market
Date range	11/3/2003 to 10/23/2013	11/3/1993 to 10/23/2013	1/3/1950 to 10/23/2013	1/4/2010 to 10/23/2013
Annual returns	6.6%	11.7%	6.2%	9.1%
Volatility	13.1%	13.5%	11.5%	16.2%
Maximum drawdown	-19.4%	-19.4%	-35.3%	-19.4%
Maximum drawdown trough date	10/3/2011	10/3/2011	11/16/1988	10/3/2011
Sharpe ratio (@ 1% risk free rate)	0.4	0.8	0.4	0.5

Chart 3: Growth of \$1000 Invested on 3 January 1950 through 23 October 2013



Note that there is no reference in this discussion to in-sample versus out-of-sample data. The strategy attempts to capture momentum, it is not designed against a particular reference frame. The concepts of in-sample and out-of-sample testing periods are not relevant here. Also note that there are more available parameters in this strategy

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that we ignored – we previously referred to these implicit parameters, than there are explicit controlling parameters. We may have tested this approach with hundreds of equities in the universe; with the inclusion of various types of stop losses; or by including exogenous economic data. None of that impacts the likely return profile of this specific strategy over long periods.

All we have to work with here are daily prices across a multitude of market cycles representing historically different market conditions. Testing the strategy across a particular time frame, for instance during the recent bull market that started 4 years ago, merely confirms what we already expect, namely that the returns over that period should be higher, and the maximum loss should be shallower, than over multiple market cycles. At best, testing over various time periods confirms that the algorithm seems bug-free, and gives us a numerical range for a behavior that we expect.

These comments apply to most strategies that rely on an economic event or investor behavior, such as many arbitrage strategies for instance. To the extent that these events or behaviors exist in the absolute, the objective of a backtest is to quantify maximum expected losses, the patterns of returns and normal losses, as well as to describe the triggers that will indicate when the arbitrage or targeted behavior will be deemed to have disappeared from the markets, at which point the strategy would need to be retired or put in abeyance until the behavior recurs. Using the longest available set of data is the most meaningful way to achieve this.

In this context what should we make of the momentum strategy backtest in this section? Table 3 indicates that we should expect to make about 6% over long periods, and potentially lose over 30% at some point. Chart 3 demonstrates that the strategy can make no money for very long periods. The portfolio's net asset value was the same in January 1962 as it was in mid-1980, with not much volatility in-between. So while the momentum effect is well documented, this implementation is unlikely to attract investors.

REVIEWING BACKTESTS FOR STATISTICAL STRATEGIES

Let's now turn to the type of investment strategies exemplified by our first example in chart 1. These strategies are simply sets of mathematical rules that produce good historical results. They don't necessarily try to capture an economic factor. They don't need to be particularly complex, as shown by our example that contained only four parameters and worked well for fifty years.

These are best described as statistical strategies. They rely on fixed mathematical rules rather than market insights. Numerous futures trading strategies fall in this category. A typical set of rule could be to buy a security when its short term momentum accelerates over a longer term of momentum, and sell when the reverse occurs.

The name of the game with these strategies is *caveat emptor*. Our approach in this paper has been to eliminate backtests through commonsensical considerations that most investor can evaluate. In the case of statistical strategies, the information asymmetry between the product sponsor and the average investor is so large that the latter is unlikely to even know what questions to ask. We are here in a completely different place than when considering economically-rooted strategies.

It is rarely possible to understand why statistical strategies should work in the future unless we assume that the majority of market participants use similar trading rules, creating a self-fulfilling feedback mechanism. Some charting techniques such as the use of Fibonacci ratios potentially fall in that category.

Consequently investors should approach a backtest with one main objective: to try and understand under which market conditions the strategy is likely to work well, and when it will not work.

An experienced and credible product sponsor understands that a statistical strategy, just like all other investment strategy ever devised, will not work well under all market conditions. A sure sign that a backtest is somehow overfitting data and that the sponsor is either too inexperienced to be credible or not entirely straightforward, is a presentation of results where all is well most of the time, or where the sponsor is unable to articulate under which market conditions the strategy will suffer. Such backtests are not worthy of any consideration.

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Having reached this point in our backtest review, we now need to differentiate between high frequency strategies and those based on daily price data that trade less frequently. Dacorogna et al.³ note that the number of observations available in a single day of tick-by-tick data, the type of data normally collected and analyzed by high frequency trading strategies, is equivalent to 30 years of daily observations. In informal conversations, high frequency traders have mentioned to your author that they regularly collect up to 60,000 individual pieces of data each day on each market security of interest. In contrast daily data consists of four price points – Open, high, Low and Close –, as well as volume information which is not usually a key driver in an algorithm.

Because of the amount of data needed to be collected and analyzed, high-frequency trading is a business with high barriers to entry. It requires a considerable investment in computer systems and software development; a highly qualified team that includes computer scientists, mathematicians and statisticians; as well as marketing, compliance and other market functions that are needed to actually operate such a company. These teams will have available to them enormous amounts of data across thousands of securities and thousands of past trading days. They are in a position to apply statistical analysis within the rules of the art. They may pick a particular date range such as a week or several months to design a strategy around a trade idea using hundreds of thousands of data points. The strategy can then be tested across a vast number of out-of-sample time periods. It can also be tested in real-time, building a meaningful body of actual market results in a matter of a few weeks or months.

High frequency trading requires such infrastructure and highly trained specialists that the management team are in a position to present investors with well thought-out statistical analysis of the market behavior it is proposing to benefit from. Whether the team will be comfortable sharing details of its strategy is a different matter.

In contrast, low-frequency statistical strategies, which are most of the strategies in use today and those we focus on in this paper, are based on daily price data. Much of this daily data is available for free at least for stocks and major market indices. Collecting ten years of daily data on a stock generates about 10,000 pieces of data – that is 4 pieces of data per stock multiplied by 10 years and 252 trading days per year. Ten years of daily data for one hundred stocks involve about one million data elements which today are easily managed with one excel spreadsheet and a five hundred dollar laptop. As a result, building a low-frequency statistical strategy backtest is a low barrier-to-entry affair. The screening approach outlined above will allow us to avoid spending time reviewing schemes concocted by quantitatively minded teams that lack credibility for one reason or another.

When reviewing low-frequency backtests that pass our selection so far, it is critical to keep in mind is that they all suffer from fundamental and incontrovertible data issues. This is because the nature of the capital markets constantly changes.

Firstly, consider that prior to the 1990's capital markets technology was primitive relative to today. There is no reason to believe that a purely statistical relationship that holds today but is not anchored into an economic phenomenon should have held when price information was hard to come by and act on.

Most product sponsors will not go through the trouble of analyzing their strategy over time periods prior to the mid 1990's for these reasons. As of 2013, this leaves a little less than twenty years of daily data available for backtest development. Considering that an equity market cycle typically lasts about seven years, we can at best expect to capture two market cycles in a backtest: The technology bubble that started in the mid-1990's and burst in April 2000; and the real estate bubble that burst in 2008, from which a number of areas of the capital markets have yet to fully recover.

Secondly, interest rates have been broadly falling between 1982 and 2012, but are now flat and may have started on secular rise. This will lead to entirely different relationships between asset classes as investors re-evaluate basic assumptions about the safety of bonds. It is difficult to overstate the impact the interest rate environment has on all investment strategies, even those that do not invest in bonds. Interest rates inform investors risk appetite and are a

³ Dacorogna, M.M., R. Gençay, U.A. Müller, R. Olsen and O.V. Pictet, 2001. *An Introduction to High-Frequency Finance*. Academic Press: San Diego, CA.

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significant, although often unseen, driver of performance in equities and other areas of the capital markets.

A complete backtest for a low-frequency statistically-based strategy should therefore include testing periods across both rising and falling interest rate environments. While capital markets price data exists for periods prior to the 1980's when interest rates were rising, it is questionable whether that data represents market action that is relevant today as we mentioned above.

As we head into a period of flat to rising interest rates, the only data available to backtest investment strategies covers a twenty-year period when rates have been falling.

Much ink has flowed on the issue of whether low-frequency strategy backtests use statistically sound data sampling methods. Statisticians and academics have pointed out that many if not most backtests seem faulty for not properly carrying out in-sample and out-of-sample analyses⁴.

From a practical point of view, we suggest that in-sample data is really all that is available to low-frequency algorithms. Credible market data encompasses only 5,000 trading days across the past twenty years, covering only two equity market cycles and less than half of a still-unfolding interest rate cycle. A strategy that trades once a month will use these 5,000 days to inform 240 trading decision points. In the absolute, these numbers are small. We believe that the existing relevant data for low-frequency backtests is simply not long enough to enable a statistical analysis that can withstand academic scrutiny as is the case for high frequency traders. Sufficient relevant data may be available to backtest low-frequency strategies with proper statistical methods in twenty or thirty years from now, when at least one whole interest-rate cycle should be available. For today, we believe that no amount of statistical wizardry can make-up for the fundamental shortcomings of the available data. Absent a statistically robust data set, sound judgment informed by experience is required to imagine and investigate what could happen with a particular strategy. This is not a road easily travelled by the uninitiated.

For experienced investors, confidence in a low-frequency statistical strategy is not necessarily increased by a backtest that purports to offer in-sample and out-of-sample analysis over a three, five or ten year period. While this may mechanically check some boxes of the statistics rule-book, one should always keep in mind that the data sandbox available to such a strategy as of 2013 is too small, no ifs and buts. The practical question is less whether such a backtest is overfitting an already insufficient data set, it is rather whether meaningful market regimes over the past fifteen to twenty years can be isolated and analyzed to develop intuition into when the strategy is likely to work, when it will not, the impact of potentially rising interest rates, and the pattern of major drawdowns investors should expect.

To put it another way, a strategy based on statistical relationships is best viewed as a black box that the investor cannot open. Besides the need to test the credibility of the product sponsor and to check that driving parameters are few and reasonably calibrated, there is little other practical use in an investor trying to find out what exactly is in the box. That is because the box simply contains a collection of rules. Spending time going through each rule is not as productive as testing the strategy to understand its behavior. The most important task is to travel in time with that black box, "plug it" into the markets at various times, and see how it behaves. Practically speaking, this means that for targeted periods selected by the investor, the sponsor should produce risk and performance statistics based on daily results, potentially benchmarked against some reference market index.

In that context it is best to target highly sensitive times period: May and June 2013 when comments by the Federal Reserve chairman roiled both equity and bond markets; October 2011 when the U.S. equity markets dropped 10% by Thanksgiving; May 6th 2010 the day of the Flash Crash; February through March 2009 when the U.S. equity markets bottomed out in a sharp U-Turn; September and October 2008 when Lehman Brothers declared bankruptcy; 2003 when equity markets started their recovery from the technology bubble burst; 2002, a full-fledged equity bear market; April 2000 when the technology bubble burst; August 1998 during the Russian default crisis; July 1997 during Asian currency crisis. The past fifteen to twenty years are rich in pivotal points that are worthwhile focusing on when

⁴ D. Bailey, J. Borwein, M. Lopez de Prado, Q. Zhu, 2013. Pseudo-Mathematics and Financial Charlatanism : *The Effects of backtest Overfitting on Out-of-Sample Performance*. Social Science Research Network.

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reviewing backtests.

This analysis usually results once again in a friendly tug of war between the product sponsor and the investor. The sponsor is keen to demonstrate and talk about periods when returns were good. The wise investor is mostly interested in periods when returns were poor or market crises developed. The objective is to try and understand these market regimes, as well as how risk is controlled when they come about. If risk is controlled effectively, returns over time will often take care of themselves.

The issues of how the strategy was initially developed, what time periods were used to define then refine risk and asset allocation parameters, and generally what statistical methods were employed by the sponsor form a natural part of the conversation around risk-control.

CONCLUDING THOUGHTS

Backtests are not all born equal. Strategies that try to capture demonstrable economic phenomena or to capitalize on certain investor behaviors have a reason of being that transcends a particular implementation proposed by a product sponsor. High-frequency strategies typically trade on such small time-scales that they have sufficient data sample to base their approach on sound statistical methods, perhaps in addition to insights in market structure or investor behavior.

In contrast, low-frequency strategies that are based purely on mathematical rules are de facto handicapped by a lack of sufficient market data. The relevant usable market data dates back to the mid 1990's, while some important asset classes such as bonds operate on cycles that exceed that range. Fortunately, an investor that cares to expend some effort in reviewing a backtested strategy can call on a number of non-technical, commonsensical tools to help select promising backtests. Discussion with the backtest sponsor about markets traded, operational requirements, strengths and weaknesses of the general investment style, team experience in dealing with severe drawdowns, number and broad settings of parameters used, form a myriad indicators that can help avoid poorly thought-out strategies. For those backtests that survive this screening, a focus on how the strategy controls risk during highly sensitive times in the markets will include a review of the sponsor's use of statistical methods. Investors do well to keep in mind that following the rule-book of statistical analysis does not make a low-frequency strategy especially likely to be successful in the future. The available dataset is too short to demonstrate such strategy's behavior across future market regimes where much price behavior is likely to change significantly from the past twenty years under the impact of new trends in interest rates.

APPENDIX

INVESTMENT STYLES EXPLAINED

There are many different ways to classify the investment strategies of money managers. Moreover, some managers combine several strategies in what is often referred to as “multi-strategy funds”. The list below illustrates the four major categories and sub-categories of investment strategies besides long-only strategies.

I. RELATIVE VALUE

This style involves the exploitation of price dislocations within different securities of either the same issuer or of issuers with similar fundamental characteristics. Often, it is the optionality that may be present in select securities, particularly convertible bonds, that is the focus. Typical strategies include convertible bond arbitrage, credit arbitrage and derivatives arbitrage. Yield alternative strategies also fall under this style. Leverage and market liquidity can be crucial factors

1. Convertible Arbitrage

In general, the strategy entails purchasing a convertible bond while simultaneously hedging a portion of the equity risk by selling short the underlying common stock. Certain managers may also seek to hedge interest rate exposure by selling Treasuries. The strategy generally benefits from three different sources: interest earned on the cash resulting from the short sales of equities, coupon offered by the bond component of the convertible and the so-called “gamma effect”. The last component results from the change in volatility of the underlying equity and involves frequent trading. This strategy is often leveraged in order to enhance returns.

2. Fixed Income Arbitrage

This strategy seeks profits by exploiting the pricing inefficiencies between related fixed-income securities while often neutralizing exposure to interest rate risk. This strategy is often leveraged in order to enhance returns.

3. Statistical Arbitrage

Managers using this strategy attempt to benefit from pricing inefficiencies that are identified using mathematical models. Statistical arbitrage strategies are based on the premise that prices will return to their historical norms. These strategies are often leveraged in order to enhance returns.

II. EVENT-DRIVEN

Major strategies within the event-driven style are distressed, and those driven by mergers and other corporate events, which fall under the risk arbitrage strategy. Sustained market declines and periods of unusual market volatility or illiquidity can be crucial factors, especially when dealing with securities that are not exchange-traded.

1. Merger Arbitrage

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Also known as risk arbitrage, this strategy invests in merger situations. The classic merger arbitrage strategy consists of being long on the stock of the target company while simultaneously selling short the stock of the acquiring company.

2. *Distressed Securities*

This investment strategy generally consists of buying securities of companies in bankruptcy proceedings and/or in the process of restructuring the debt portion of their balance sheets. The complexity of such operations often creates mispricing opportunities, hence high potential returns.

3. *Special Situations*

Also known as corporate life cycle, this strategy focuses on opportunities created by significant transactional events, such as division spin-offs, mergers, acquisitions, bankruptcies, reorganizations, share buybacks, and management changes.

III. EQUITY HEDGE

Equity Hedge strategies have equity market exposure and, in general, tend to have a bias toward a net long position, which often leads to higher correlation to common market indices. Managers typically run portfolios on a highly-hedged basis. Returns can be sourced from fundamental or quantitative methods, both within sectors or across sectors; however a general aim is to avoid beta exposure in the portfolio.

1. *Growth/Value/Industry/Geographical/ Capitalization*

This style accounts for the majority of the strategies used by fund managers today. This directional strategy combines both long and short positions in stocks. The net market exposure is adjusted opportunistically. The manager can diversify holdings across different industries, countries, market capitalizations, etc...

2. *Market Neutral*

This strategy is designed to exploit inefficiencies in the equity market by trying to remove the element of systematic risk while extracting the stock-specific returns. These portfolios minimize market risk by being simultaneously long and short on stocks having different characteristics.

3. *Short Sellers*

The short selling approach seeks to profit from declines in the value of stocks. The strategy consists of borrowing a stock and selling it on the market with the intention of buying it back later at a lower price. By selling the stock short, the seller receives interest on the cash proceeds resulting from the sale. If the stock advances, the short seller takes a loss when buying it back to return to the lender.

IV. MACRO

This style focuses on arbitrage-related trading in a broader range of markets than equities and/or bonds, utilizing commodities and futures as well. Investment processes can be purely model-driven or fundamental, and there is often a momentum component involved. Price trends and patterns in the futures markets are the source of opportunity in a deeply liquid market.

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1. Global Macro

Global Macro managers make in-depth analyses of macro-economic trends and formulate their investment strategy based on these, taking out positions on the fixed income, currency and equity markets through either direct investments or futures and other derivative products.

2. CTA

CTA is the acronym for Commodity Trading Advisor and is also known as Managed Futures. This strategy essentially invests in futures contracts on financial, commodity, and currency markets around the world. Trading decisions are often based on proprietary quantitative models and technical analysis. These portfolios have embedded leverage through the derivative contracts employed.

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